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Data Science
São Paulo School of Advanced Science on Learning from Data

Detection of Architectural Distortion of Breast in Digital Breast Tomosynthesis Images

PhD Thesis

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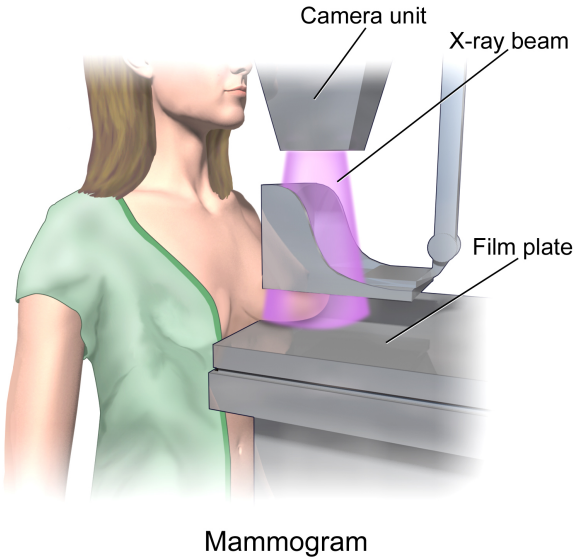
São Carlos School of Engineering (EESC)
University of São Paulo (USP)

Objective

- **Digital Breast Tomosynthesis** (DBT):



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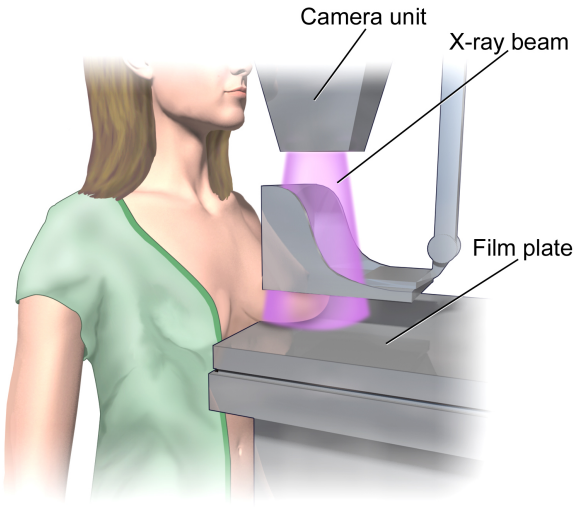
Objective

- **Digital Breast Tomosynthesis** (DBT):
  - New tool for breast cancer screening;



Mammogram

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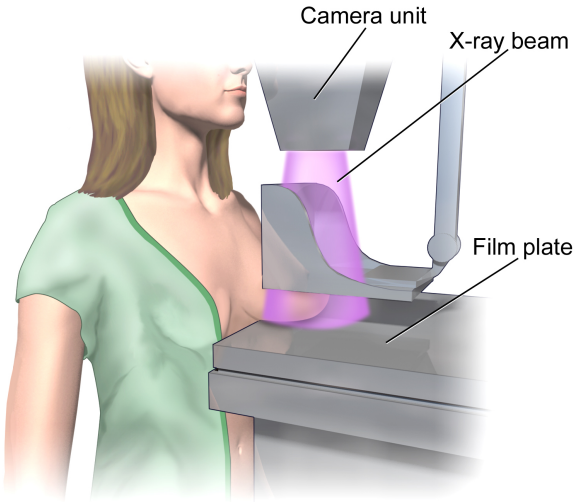
Objective

- **Digital Breast Tomosynthesis** (DBT):
  - New tool for breast cancer screening;
- Main signs of breast cancer:
  - Masses;
  - Calcifications;
  - **Architectural Distortion** (AD):



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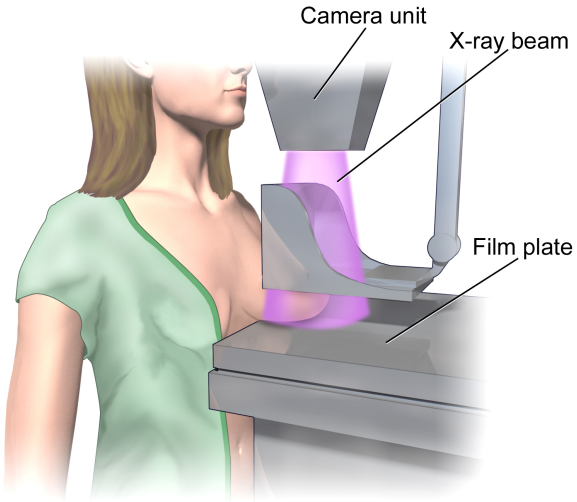

Mammogram

- Main signs of breast cancer:
  - Masses;
  - Calcifications;
  - **Architectural Distortion** (AD):
    - Can be the earliest sign of breast cancer;
    - Very difficult to be detected by radiologists;

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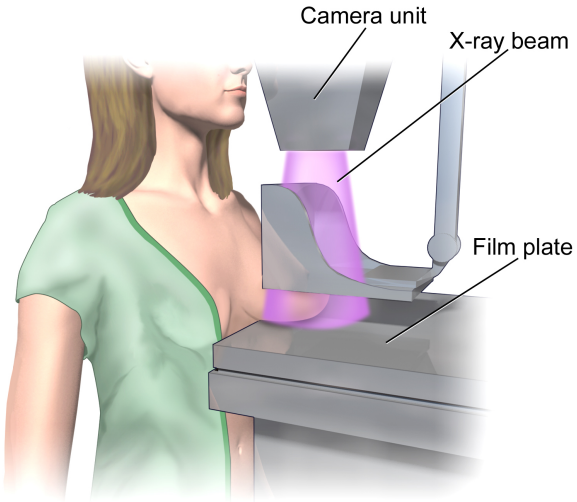

Mammogram

- **Digital Breast Tomosynthesis** (DBT):
  - New tool for breast cancer screening;
- Main signs of breast cancer:
  - Masses;
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    - Can be the earliest sign of breast cancer;
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- Proposal: **Improve the Detection of AD in DBT images**.

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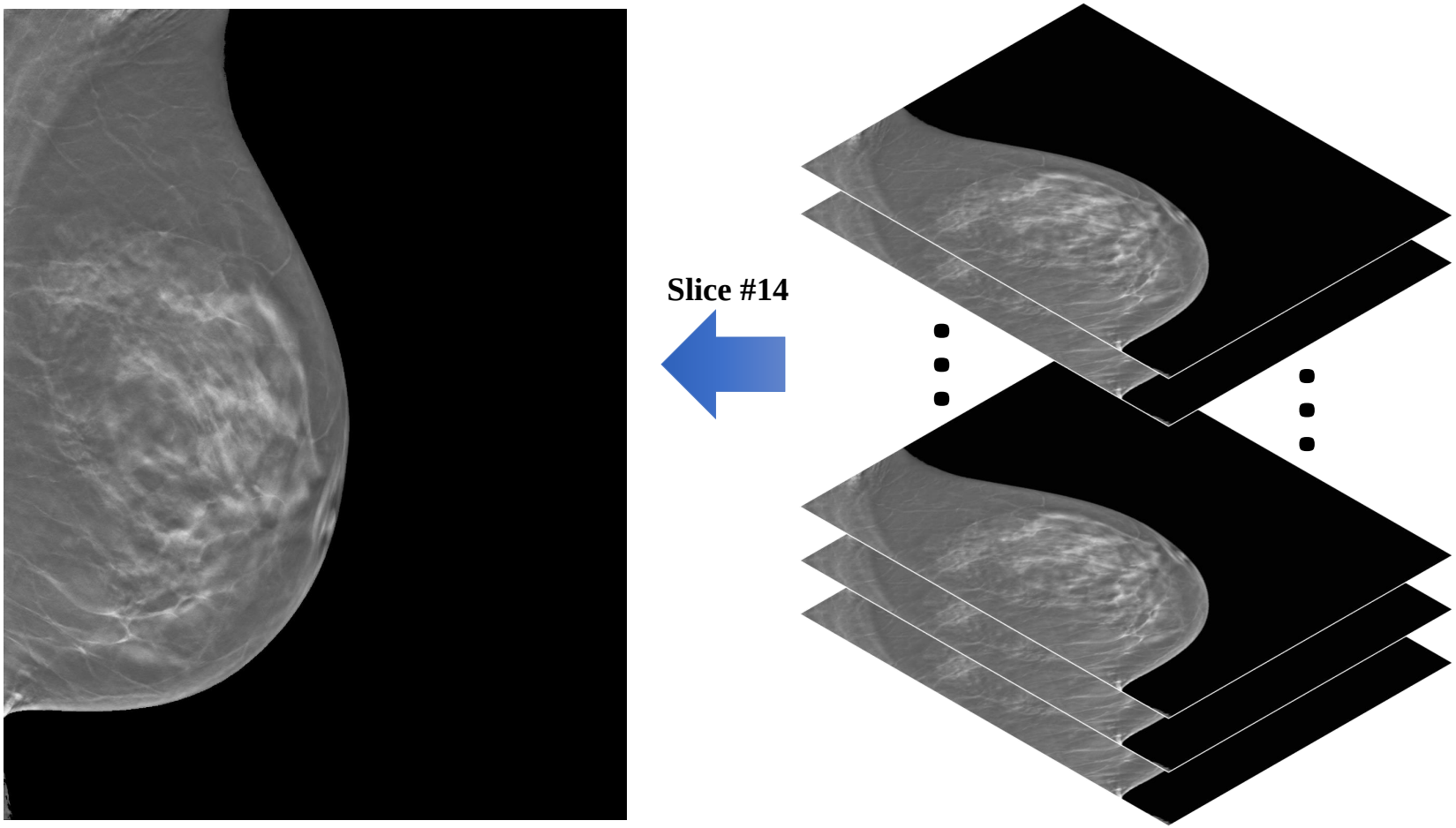
DBT Images



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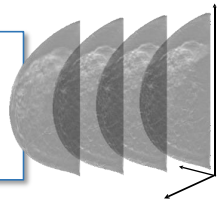
Proposed Pipeline

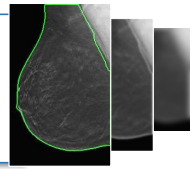
Preprocessing

- Build stack of slices;
- Contrast enhancement (LUT);



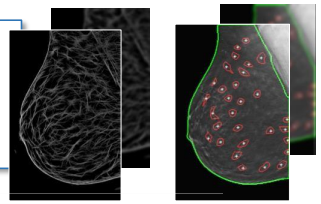
Segmentation

- Background subtraction;
- Identify pectoral muscle;
- Segment breast contour;



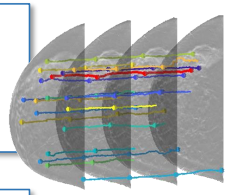
Detection of Suspicious Spots

- Gabor filtering;
- Compute Gaussian curvature;
- Centroids extraction;



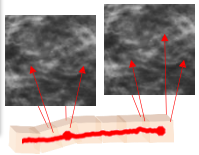
Tracking Suspicious Spots

- Construct binary maps;
- Run tracking algorithm;
- Remove spurious track;



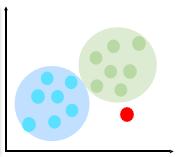
Feature Extraction

- Extract ROI stack;
- Compute ROIs features;
- Aggregate features at track level;



AD Recognition

- Remove false AD track;
- Apply LDA classifier;



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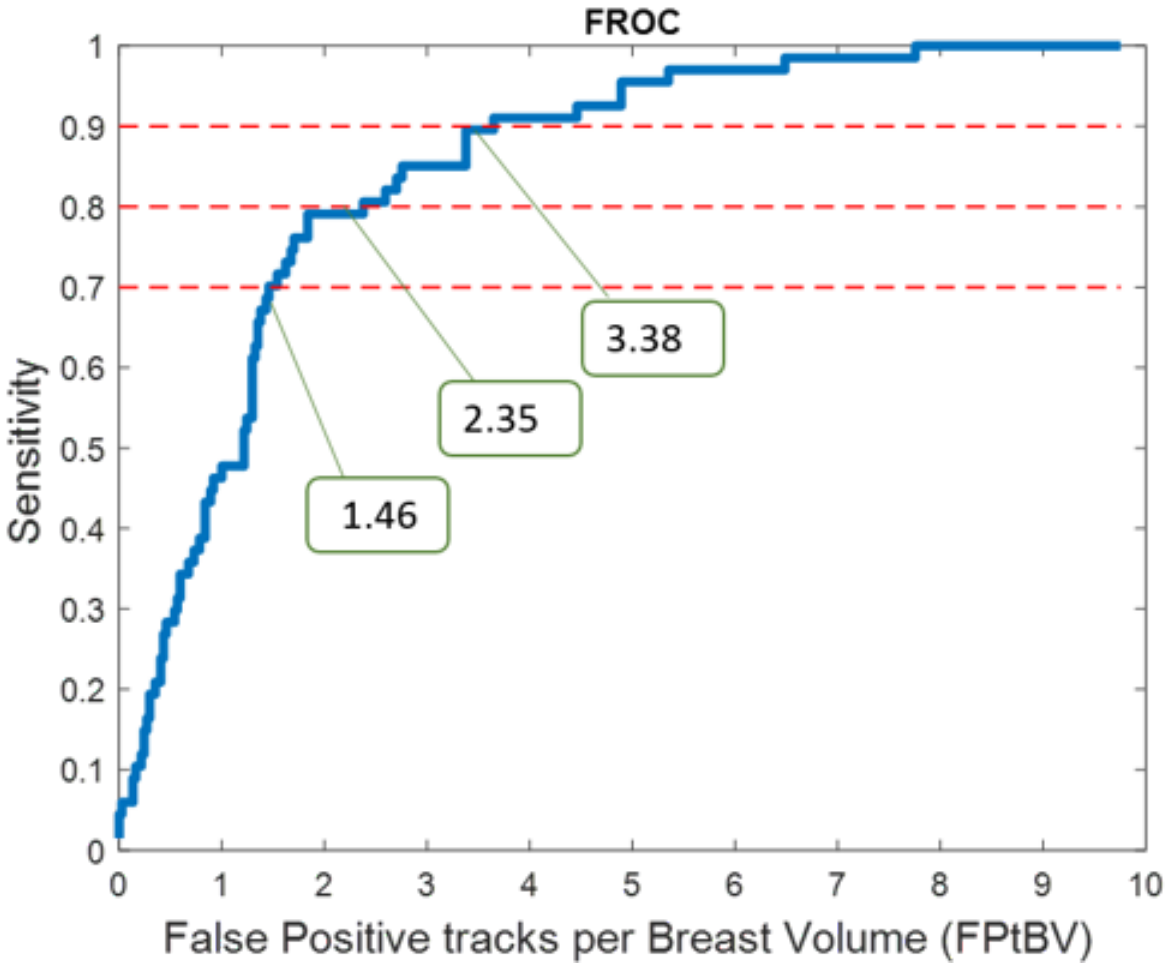
Results



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Results

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A cross-cutting approach for tracking architectural distortion locii on digital breast tomosynthesis slices

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ABSTRACT

Background and objective: Full-field digital mammography (FFDM) is the predominant breast cancer screening exam used. However, with the emergence of digital breast tomosynthesis (DBT) the radiologists could improve early recognition of breast cancer signs. In this scenario, the detection of architectural distortion (AD) is still a challenging task. ADs are very subtle contraction of the breast parenchyma that could represent the earliest manifestation of cancer, assessing at present 50% of missed cases. *Methods:* This paper proposes a new paradigm to detect AD in DBT exams by a cross-cutting approach exploiting the 3-dimensionality of the imaging modality. After locating AD candidates in each DBT slice, the suspicious spots are tracked in cross-slice direction and then characterized in terms of neighboring texture. In this approach, which mimics radiologist's scrolling down over zoomed slices, we reduce the amount of uninformative signs collected in DBT exams by preserving the large variability of AD appearance. *Results:* Using 37 sets of DBT slices containing at least one AD locus indicated by a radiologist, the proposed methodology reaches an AUC of 0.84, with only one false negative exam at sensitivity of 0.9. *Conclusions:* The results show that the proposed algorithm can be a promising tool for the automatic detection of AD locii. Future work will address the extension of the dataset of DBT slices as well the improvement of algorithm performance toward the application in the clinical practice.

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